Web Recon Automation Tool Blueprint

Purpose

Build a Linux-based reconnaissance framework that gathers information about an authorized target website from a URL and produces a consolidated report.

High-Level Workflow

Target URL → URL Normalizer → Passive Recon + Active Recon → Information Parser → Technology Detection → Endpoint Discovery → Security Configuration Analysis → Report Generator (HTML/PDF/JSON).

Core Modules

1. Target Processing (domain, IP, host, scheme, port)
2. DNS (A, AAAA, MX, TXT, SPF, DKIM, DMARC, NS, SOA, CNAME)
3. SSL/TLS (certificate, issuer, validity, cipher, TLS version, HSTS)
4. WHOIS/ASN
5. HTTP headers, cookies, redirects, server info
6. Technology detection (frontend, backend, CMS, server)
7. Security headers
8. Cookie analysis
9. robots.txt, sitemap.xml, security.txt
10. Website crawler
11. JavaScript analysis
12. Form detection
13. Authentication detection
14. API discovery
15. File upload/download discovery
16. CDN/WAF detection
17. Screenshot capture
18. AI business analyzer
19. Reporting

Recommended Tech Stack

Language: Python 3.12+
CLI: Typer
HTTP: httpx + asyncio
Browser: Playwright
HTML Parsing: BeautifulSoup4
DNS: dnspython
Reporting: Jinja2 + ReportLab/WeasyPrint
Database: SQLite
Logging: Rich + Loguru

Project Structure

reconx/
main.py
config.yaml
modules/
templates/

reports/
screenshots/
cache/
utils/

Development Roadmap

Phase 1: Core Recon
Phase 2: Web Mapping
Phase 3: AI Intelligence
Phase 4: Reporting
Phase 5: Plugin Architecture & Integrations

Expected Outputs

Business overview, infrastructure, DNS, TLS, HTTP headers, technologies, authentication, APIs, endpoints, forms, JavaScript findings, screenshots, security observations, recommendations, and HTML/PDF/JSON reports.

Important Note

Use this framework only on systems you own or are explicitly authorized to assess.